

Advanced Python Practical Answers - Corrected Version

SLIP 1

Q1) Write a program that reads all the lines from a file named "data.txt" and prints each line to the console

Answer:

```
with open("data.txt", "r") as file:
    lines = file.readlines()
```

```
for line in lines:
    print(line.strip())
```

Output:

```
Hello
Python
Programming
```

Q2) Write a Python script that creates two threads.

Answer:

```
import threading
import time
```

```
def print_numbers():
    for i in range(1, 6):
        print(i)
        time.sleep(1)
```

```
def print_letters():
    for ch in ['A', 'B', 'C', 'D', 'E']:
        print(ch)
        time.sleep(1)
```

```
t1 = threading.Thread(target=print_numbers)
t2 = threading.Thread(target=print_letters)
```

```
t1.start()
t2.start()
```

```
t1.join()
t2.join()
```

Output:

```
1
A
2
B
3
C
4
D
5
E
```

SLIP 2

Q1) Write a regular expression to validate a phone number format.

Answer:

```
import re
```

```
pattern = r"^\(\d{3}\) \d{3}-\d{3}$"
number = "(123) 456-789"
```

```
if re.match(pattern, number):
    print("Valid Number")
else:
    print("Invalid Number")
```

Output:

```
Valid Number
```

Q2) Write a Python program demonstrating Hierarchical Inheritance.

Answer:

```
class Shape:
    def area(self):
        pass
```

```
class Circle(Shape):
```

```

    def __init__(self, radius):
        self.radius = radius

    def area(self):
        return 3.14 * self.radius * self.radius

class Rectangle(Shape):
    def __init__(self, length, breadth):
        self.length = length
        self.breadth = breadth

    def area(self):
        return self.length * self.breadth

c = Circle(5)
r = Rectangle(4, 6)

print("Circle Area:", c.area())
print("Rectangle Area:", r.area())

```

Output:
Circle Area: 78.5
Rectangle Area: 24

SLIP 3

Q1) Write a Python script that opens a file named "example.txt" in write mode.

Answer:

```

file = open("example.txt", "w")
file.write("Hello, World!")
file.close()

print("File Written Successfully")

```

Output:
File Written Successfully

Q2) Write a Python program that demonstrates multiple inheritance.

Answer:

```

class Person:
    def show_person(self):
        print("Person Class")

class Employee:
    def show_employee(self):
        print("Employee Class")

class Manager(Person, Employee):
    def show_manager(self):
        print("Manager Class")

m = Manager()

m.show_person()
m.show_employee()
m.show_manager()

```

Output:
Person Class
Employee Class
Manager Class

SLIP 4

Q1) Write a program that reads all the lines from "data.txt".

Answer:

```

with open("data.txt", "r") as file:
    for line in file:
        print(line.strip())

```

Output:
Line 1
Line 2
Line 3

Q2) Write a Python GUI program to create a window and set its title.

Answer:

```

from tkinter import *
root = Tk()

```

```
root.title("Python Window")

root.mainloop()
```

Output:
Window created successfully

SLIP 5

Q1) Write a script that checks if "config.ini" exists.

Answer:

```
import os

if os.path.exists("config.ini"):
    print("File exists")
else:
    print("File does not exist")
```

Output:
File exists

Q2) Write a Python GUI program to create a window with label.

Answer:

```
from tkinter import *

root = Tk()
root.title("Tkinter Window")

label = Label(root, text="Welcome")
label.pack()

root.mainloop()
```

Output:
Window with label displayed

SLIP 6

Q1) Read a binary file and copy its content.

Answer:

```
with open("image.png", "rb") as source:
    data = source.read()

with open("copy_image.png", "wb") as target:
    target.write(data)

print("File copied successfully")
```

Output:
File copied successfully

Q2) Create a Tkinter window with title "Hello World" and size 300x200.

Answer:

```
from tkinter import *

root = Tk()
root.title("Hello World")
root.geometry("300x200")

root.mainloop()
```

Output:
Window created successfully

SLIP 7

Q1) Demonstrate regular expressions.

Answer:

```
import re

text = "My mobile number is 9876543210"
result = re.findall(r"\d+", text)

print(result)
```

Output:

```
['9876543210']
```

Q2) Create Tkinter widgets.

Answer:

```
from tkinter import *  
  
root = Tk()  
  
Label(root, text="Name").pack()  
  
Entry(root).pack()  
  
Button(root, text="Submit").pack()  
  
Text(root, height=5, width=20).pack()  
  
root.mainloop()
```

Output:

Widgets created successfully

SLIP 8

Q1) Demonstrate threading and multitasking.

Answer:

```
import threading  
  
def task1():  
    print("Task 1 Running")  
  
def task2():  
    print("Task 2 Running")  
  
t1 = threading.Thread(target=task1)  
t2 = threading.Thread(target=task2)  
  
t1.start()  
t2.start()  
  
t1.join()  
t2.join()
```

Output:

Task 1 Running
Task 2 Running

Q2) Create a Tkinter label with red color and font size 20.

Answer:

```
from tkinter import *  
  
root = Tk()  
  
label = Label(root, text="Hello, Tkinter!", fg="red", font=("Arial", 20))  
label.pack()  
  
root.mainloop()
```

Output:

Label displayed successfully

SLIP 9

Q1) Find all occurrences of "abc".

Answer:

```
import re  
  
text = "abc xyz abc pqr abc"  
  
result = re.findall("abc", text)  
  
print(result)
```

Output:

```
['abc', 'abc', 'abc']
```

Q2) Create a button that changes label text.

Answer:

```

from tkinter import *

def change_text():
    label.config(text="Button Clicked")

root = Tk()

label = Label(root, text="Hello")
label.pack()

button = Button(root, text="Click", command=change_text)
button.pack()

root.mainloop()

Output:
Label text changed successfully

```

SLIP 10

Q1) Find sequences of digits with exactly length 3.

```

Answer:
import re

text = "123 4567 789 12 345"

result = re.findall(r"\b\d{3}\b", text)

print(result)

```

```

Output:
['123', '789', '345']

```

Q2) Department class with Employee objects.

```

Answer:
class Employee:
    def __init__(self, name):
        self.name = name

class Department:
    def __init__(self):
        self.employees = []

    def add_employee(self, emp):
        self.employees.append(emp)

    def display_employees(self):
        for emp in self.employees:
            print(emp.name)

```

```

e1 = Employee("Rohan")
e2 = Employee("Amit")

```

```

d = Department()

d.add_employee(e1)
d.add_employee(e2)

d.display_employees()

```

```

Output:
Rohan
Amit

```

SLIP 11

Q1) Match string with a followed by three b.

```

Answer:
import re

text = "abbb"

if re.fullmatch(r"abbb", text):
    print("Matched")
else:
    print("Not Matched")

```

```

Output:
Matched

```

Q2) Create Employee class with class variable.

Answer:

```

class Employee:
    company_name = "ABC Ltd"

    def __init__(self, name, salary):
        self.name = name
        self.salary = salary

    def display(self):
        print(self.name, self.salary, Employee.company_name)

    @classmethod
    def change_company(cls, new_name):
        cls.company_name = new_name

e1 = Employee("Rohan", 50000)
e1.display()

Employee.change_company("XYZ Ltd")

e2 = Employee("Amit", 60000)
e2.display()

Output:
Rohan 50000 ABC Ltd
Amit 60000 XYZ Ltd

```

SLIP 12

Q1) Read all lines from "data.txt".

```

Answer:
with open("data.txt", "r") as file:
    for line in file:
        print(line.strip())

```

```

Output:
Python
Java
C++

```

Q2) Demonstrate instance and class namespace.

```

Answer:
class Car:
    wheels = 4

    def __init__(self, make, model):
        self.make = make
        self.model = model

c1 = Car("Toyota", "Innova")
c2 = Car("Honda", "City")

print(c1.make, c1.model, c1.wheels)
print(c2.make, c2.model, c2.wheels)

```

```

Output:
Toyota Innova 4
Honda City 4

```

SLIP 13

Q1) Read binary file and copy content.

```

Answer:
with open("image.png", "rb") as file1:
    data = file1.read()

with open("copy_image.png", "wb") as file2:
    file2.write(data)

print("File copied")

```

```

Output:
File copied

```

Q2) Create Person class with display method.

```

Answer:
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

```

```

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)

p = Person("Rohan", 20)

p.display()

```

Output:
Name: Rohan
Age: 20

SLIP 14

Q1) Find all occurrences of sequence "abc".

Answer:

```

import re

text = "abc test abc demo"

result = re.findall("abc", text)

print(result)

```

Output:
['abc', 'abc']

Q2) Create Student and Course classes.

Answer:

```

class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def display(self):
        print(self.name, self.age)

class Course:
    def __init__(self, course_name):
        self.course_name = course_name
        self.students = []

    def add_student(self, student):
        self.students.append(student)

    def display_students(self):
        for student in self.students:
            student.display()

s1 = Student("Rohan", 20)
s2 = Student("Amit", 21)

c = Course("Python")

c.add_student(s1)
c.add_student(s2)

c.display_students()

```

Output:
Rohan 20
Amit 21

SLIP 15

Q1) Match a string that has an a followed by zero or more b's.

Answer:

```

import re

text = "abbb"

if re.fullmatch(r"ab*", text):
    print("Matched")
else:
    print("Not Matched")

```

Output:
Matched

Q2) Create base class Vehicle and derived class Car.

Answer:

```
class Vehicle:
    def __init__(self, speed):
        self.speed = speed

    def drive(self):
        print("Vehicle is moving")

class Car(Vehicle):
    def __init__(self, speed, fuel_type):
        super().__init__(speed)
        self.fuel_type = fuel_type

c = Car(120, "Petrol")

c.drive()
print("Speed:", c.speed)
print("Fuel Type:", c.fuel_type)
```

Output:

```
Vehicle is moving
Speed: 120
Fuel Type: Petrol
```